

Read and structure expressions

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: Describe the calculation $48 - 6 \times 5$ without solving it. Copy or complete the first step.

2. Which operation happens first in $72 \div 8 + 4$?

3. Write an expression for subtract 7 from the product of 9 and 6.

4. Miro says: Miro reads every expression strictly from left to right. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Describe the calculation $48 - 6 \times 5$ without solving it. Copy or complete the first step.

Answer Multiply 6 by 5, then subtract the product from 48.

2. Which operation happens first in $72 \div 8 + 4$?

Answer Division.

3. Write an expression for subtract 7 from the product of 9 and 6.

Answer $9 \times 6 - 7$.

4. Miro says: Miro reads every expression strictly from left to right. Is this correct? Explain.

Answer The structure of the expression determines the order, not reading direction alone.

Read and structure expressions

Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. Describe the calculation $48 - 6 \times 5$ without solving it.

6. Check this result using an inverse, estimate or known fact: Write an expression for subtract 7 from the product of 9 and 6.

2. Which operation happens first in $72 \div 8 + 4$?

7. Insert brackets into $8 + 4 \times 3$ to make the answer 36.

3. Write an expression for subtract 7 from the product of 9 and 6.

8. Identify and correct this misconception: Miro reads every expression strictly from left to right.

4. Solve this independently and show every stage: Describe the calculation $48 - 6 \times 5$ without solving it.

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: Which operation happens first in $72 \div 8 + 4$?

10. Write and solve a similar question to: Write an expression for subtract 7 from the product of 9 and 6.

Teacher answers and checking prompts

1. Describe the calculation $48 - 6 \times 5$ without solving it.
Answer Multiply 6 by 5, then subtract the product from 48.
2. Which operation happens first in $72 \div 8 + 4$?
Answer Division.
3. Write an expression for subtract 7 from the product of 9 and 6.
Answer $9 \times 6 - 7$.
4. Solve this independently and show every stage: Describe the calculation $48 - 6 \times 5$ without solving it.
Answer Multiply 6 by 5, then subtract the product from 48.
5. Use a second method or representation for: Which operation happens first in $72 \div 8 + 4$?
Answer Expected result: Division. Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: Write an expression for subtract 7 from the product of 9 and 6.
Answer Expected result: $9 \times 6 - 7$. A valid check must be shown.
7. Insert brackets into $8 + 4 \times 3$ to make the answer 36.
Answer $(8 + 4) \times 3 = 36$.
8. Identify and correct this misconception: Miro reads every expression strictly from left to right.
Answer The structure of the expression determines the order, not reading direction alone.
9. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: Write an expression for subtract 7 from the product of 9 and 6.
Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Read and structure expressions

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. Insert brackets into $8 + 4 \times 3$ to make the answer 36.

6. Create a new example that tests the same idea as: Write an expression for subtract 7 from the product of 9 and 6.

2. Prove the answer to this question, rather than only calculating it: Describe the calculation $48 - 6 \times 5$ without solving it.

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: Which operation happens first in $72 / 8 + 4$?

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: $9 \times 6 - 7$.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro reads every expression strictly from left to right.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Insert brackets into $8 + 4 \times 3$ to make the answer 36.
Answer $(8 + 4) \times 3 = 36$.
2. Prove the answer to this question, rather than only calculating it: Describe the calculation $48 - 6 \times 5$ without solving it.
Answer Expected result: Multiply 6 by 5, then subtract the product from 48. Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: Which operation happens first in $72 / 8 + 4$?
Answer Expected result: Division. Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: $9 \times 6 - 7$.
Answer One valid reconstruction is: Write an expression for subtract 7 from the product of 9 and 6. Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro reads every expression strictly from left to right.
Answer The structure of the expression determines the order, not reading direction alone.
6. Create a new example that tests the same idea as: Write an expression for subtract 7 from the product of 9 and 6.
Answer Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
Answer Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.